Ready to spend: run your first ad as an experiment

What your testers showed you. What $100 can answer. Proof that your tracking works.

Class 9 · Friday, November 6, 2026 · 1:10 to 3:40 PM · Pupin 420

What we're doing today

1:10 PM

Check-in

News, the leaderboard, your endpoint, presentation slots.

1:34 PM

Debrief

What your testers did, and where they got stuck.

1:49 PM

Ad experiment

Your ad as an experiment, studio time, then analytics.

3:32 PM

Homework

What is due Friday, November 13.

The rest of the semester, 1 of 2

Date	Class	Topic	Homework due
Today	Class 9	Paid ads as an experiment	
Nov 13	Class 10	Growth: channels and cold start	Usability tests, ad card, ad live
Nov 20	Class 11	Monetization and business models	Ad test results
Nov 27	No class	Thanksgiving	

The rest of the semester, 2 of 2

Date	Class	Topic	Homework due
Dec 4	Class 12	Startup 101 and pitch decks	Iterate, then freeze
Dec 11	Class 13	Pitch studio time	Pitch deck draft 1
Dec 18	Demo Day	Final pitches and live demos	Final deck

What you work on next, and why

Next class, Nov 13

- Class 10: Growth: channels and cold start
- Usability tests, ad card, ad live
- Why it matters: A product with no users gives people no reason to join. You need a plan for getting the first ones.

After that, Nov 20

- Class 11: Monetization and business models
- Ad test results
- Why it matters: How you will make money shapes what you build next. Unit economics shows whether the math works.

AI news this week

Wispr Flow grew by paying creators to make ads

- About 50 creators make 500 to 600 paid ads a month
- Creators earn a cut of the ad spend
- Raised $280M at a $2B valuation in August
- Some users say dictation quality has dropped lately

Lenny's Newsletter, Nov 1 · TechCrunch, Aug 17

A solo founder's rule: never check a stand-in

- Rebuilt his whole platform alone with Claude Code
- Tests passed for weeks; the real database never had the record
- A nightly report said work was done when nothing ran
- Cheap code made unclear instructions expensive

David DiCarlo, The AI Collective, Nov 3

Elena Verna: AI may end the hunt for features

- Users state a goal; the software picks the features
- New problems: trust, showing what the AI did, undo
- Engagement may fall even as customer value rises
- A blank prompt box doesn't show what you can do

Elena Verna, Lenny's Newsletter, Nov 5

Why the number matters

Your grade

10%

User growth is a tenth of the course grade

Per point

10

Real users earn 1 point

Full credit

100

Real users or more earns every point

Automate the leaderboard

- I don't want to ask you for numbers every class.
- **You build the API endpoint.**
- I pull your numbers automatically before each class. No manual reporting.

Your API endpoint

Build a public endpoint that returns JSON:

```
GET https://your-app.com/api/metrics
{
  "signups": 42,
  "active_users": 15,
  "waitlist": 147,
  "page_views": 820
}
```

What each metric means

- **signups:** registered accounts in your database
- **active_users:** did something meaningful (you define what)
- **waitlist:** email signups before the product is live
- **page_views:** from GA4 or your analytics tool

Return 0 for any metric you don't have yet. That's fine.

Put your URL in GitHub

One file, one URL. Commit and push. I'll grab it automatically.

```
teams/your-team-name/api-metrics.md
https://your-app.com/api/metrics
```

Endpoint requirements

- **Public:** no auth required, it's just aggregate counts
- **Returns JSON** with the four metric fields
- **Always up:** if it's down when I check, you get zeroes on the board
- **Counts real users** from your database, not hardcoded

What I noticed in your weekly updates

- Two teams sent one early. Thank you.
- The and why is the useful part.
- Read your AI draft before you submit it.
- Due before class every Friday, starting November 13.

The weekly update template

teams/your-team-name/weekly_update_YYYY-MM-DD.md, half a page

```
Weekly Update: [Team name], [Date]

Accountability: what we built last week and why
Planning: what we're building this week and why
Reporting: users total, change since last week
Growth: what changed in our demand strategy
Issues: biggest blocker (or "None")
```

Two slots left. Claim them now.

Date	Class	Slot 1	Slot 2	Slot 3
Nov 13	Class 10	AstraPath	Settlewise	
Nov 20	Class 11	AIDI	Black Panthers	
Dec 4	Class 12	5 PPL	Aditya	ZeroOne
Dec 11	Class 13	Open	Open	

Still need a slot: Five Girls, The Babysitters.

Testing debrief

Part 2

Sixty to ninety seconds per team

- One thing a tester did that you didn't **expect**
- Where they got **stuck** on the way to your aha moment

Tested real people this week? You go first.

"She clicked the X three times" is data.

"She said it was confusing" is not.

Where testers got stuck

Fill in live as each team reports.

Team	Did something unexpected	Got stuck at

Part 3

Ready to spend: your ad is an experiment

Where you should be by now

1. Understand problems by talking to people
2. See what people are saying unprompted
3. Create AI personas to embed what you learned
4. Create a website and landing pages
5. Drive early traffic and create a growth plan
6. Run usability tests on onboarding and the aha moment

All of that sets you up to spend money with confidence, and not waste it.

Ready to spend

So now you spend money.

$100, one question, and a bar you write down first.

Activation first

Fix onboarding before scaling traffic.

Otherwise you're filling a leaky bucket. You can't retain someone who never got the point.

An experiment changes one thing on purpose and compares.

Everything else is an observation.

Before you spend, write down

1. Your hypothesis, with a number in it
2. The number that means go
3. The number that means stop
4. The date you will look

The rule exists before the data does.

it is always needful to forecast all possible results of the experiment, and to have decided without ambiguity what interpretation shall be placed upon each one of them

R. A. Fisher, The Design of Experiments, 2nd ed., 1937, section 6

Fisher, The Design of Experiments (scan)

Authors must decide the rule for terminating data collection before data collection begins and report this rule in the article

Simmons, Nelson and Simonsohn, "False-Positive Psychology," Psychological Science, 2011

Psychological Science 22(11), Table 2, p. 1362

Change one thing because $100 buys one question.

Not because one variable is the scientific ideal. Fisher said so himself.

1 in 3

of tested ideas at Microsoft, roughly, improved their target metric

Kohavi et al., Online Experimentation at Microsoft, 2009

Elsewhere, by Kohavi's count

His own compilation, not rates the companies published.

Bing

15%

of tested ideas succeeded

Google Ads

10%

of tested ideas succeeded

Airbnb Search

8%

of tested ideas succeeded

Kohavi, Deng and Vermeer, KDD 2022, Table 2

A flat result is the normal result.

Plan for it before you launch.

What experienced teams do

01	**Rule first**	Write what counts as success before the test runs.
02	**One number**	One decision metric, plus a few guardrails.
03	**A full week**	At least seven days, because behavior differs by weekday.
04	**Replicate**	A surprising win earns a second test, not an announcement.

Any figure that looks interesting or different is usually wrong

Twyman's law, as printed in Kohavi, Tang and Xu, Trustworthy Online Controlled Experiments, 2020

Kohavi, Tang and Xu, Ch. 3, p. 39

If it looks amazing, check the tracking first.

A 20% signup rate from 30 clicks gets a tracking check before it gets an announcement.

What transfers to a team with no traffic

✓ The habits

- Write the rule first
- One number decides
- Replicate a surprising win

✕ The statistics

- They run on huge traffic
- About 121,600 users per variant
- Kohavi distrusts tests with 2,000 users

You copy how they think, not how they measure.

With tens of users, there's nothing to A/B test.

Early founders ran existence tests instead:
will anyone take a costly step at all?

Existence tests, in the founders' own accounts

Zappos	Photographed a store's shoes; offered to buy at full price if anyone ordered. A few did.
IMVU	Zero debate avatars sold. New prices, copy, and pages didn't change that.
Buffer	A pricing page before the email form. A small number clicked paid plans.

Existence tests, continued

Dropbox	A demo video on Digg. The waitlist went from 5,000 to 75,000 in one day.
Tim Ferriss	Tested book titles on AdWords, about $200 over about a week. Picked the top click rate.

Big answers or zero. That's the shape $100 can see.

The more the action costs the visitor, the less it fools you.

A pre-order or a price click
tells you more than an email.

Shortcuts that fool you

- Comparing before and after
- Re-cutting the data after you see it
- Counting signups from friends
- Calling a zero on a handful of clicks

Ask of each one: could it have come out against you?

What $100 buys

Meta

166

clicks, at about $0.60 a click

Google Search

18

clicks, at about $5.42 a click

Vendor medians: WordStream Facebook 2026 · WordStream Google 2026

What 166 clicks can and cannot tell you

Question	166 clicks?
A weak page (2%) or a good one (8%)?	Yes
Does the whole path work, ad to signup?	Yes
Headline A or headline B?	No
5% or 7.5%?	No: needs 1,273 clicks per page

Sample sizes: Evan Miller's calculator; scripts in the report folder

Clicks you need to tell two pages apart

One page	The other	Clicks per page
5%	10%	336
3%	6%	574
10%	20%	157
2%	4%	872

Halve the gap you want to see, and you need about four times the clicks.

What $100 can answer

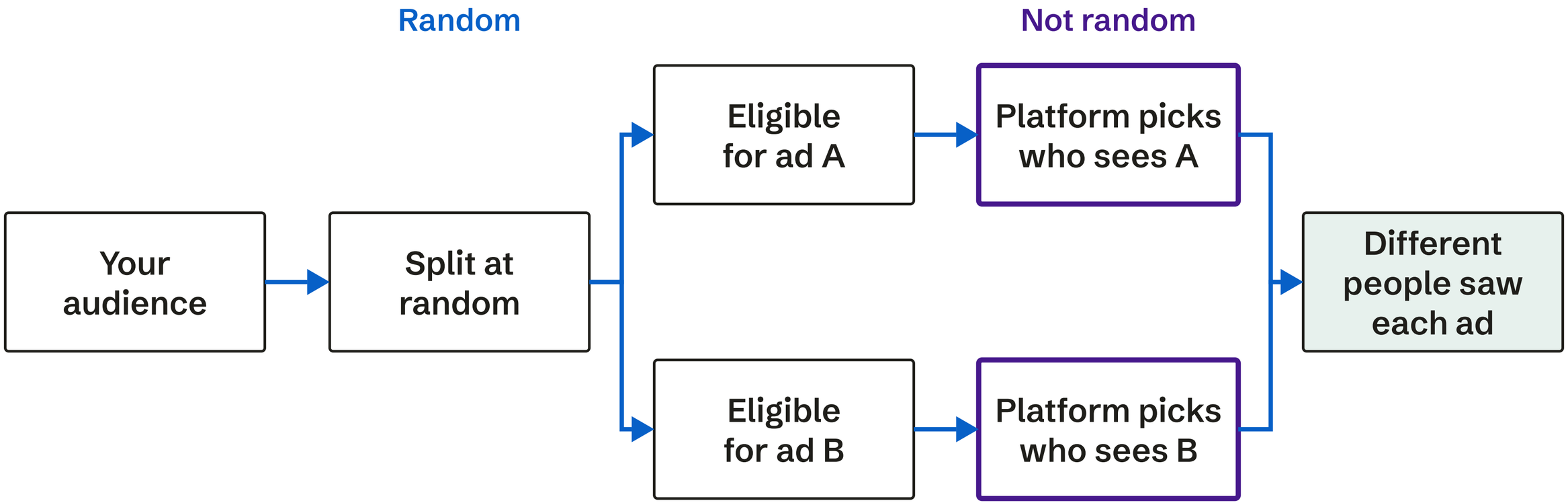
Telling 5% from 5.5% takes 30,245 clicks per page.

About $36,294 on Meta, for both pages.

Zero signups from 18 clicks still allows a true rate up to 17.6%.

Zero from 166 clicks rules out anything above about 2.3%.

The split is random. Who sees each ad is not.



"Ad A beat ad B" can mean "ad A found better people."

Four research groups show this,
including researchers who work at Meta.

Four groups, the same finding

Ali, 2019	Same targeting: a bodybuilding ad reached over 80% men, cosmetics over 90% women.
Eckles, 2018	Each version reached a different age and gender mix.
Braun, 2025	14 recruiting ads in one A/B test reached different shares of women.
Burtch, 2025	Mostly Meta researchers. Big audience gaps in 22% of A/B comparisons.

Ad A vs ad B

65%

confidence is all Meta and Reddit need to call a winner

Meta, About confidence · Reddit, July 2026

Ad A vs ad B

Two identical ads clear that bar roughly 4 to 7 times in 10.

A green trophy in Ads Manager is not evidence at $100.

A result too small to call

- Page A: 4 signups from 83 clicks (4.8%)
- Page B: 7 signups from 83 clicks (8.4%)
- B looks 75% better.

A gap that large shows up about 35% of the time between identical pages.

The real test tools cost more than you have

Tool	Needs	Against your $100
Reddit Split Testing	$1,000 a day minimum	10 times, per day
Google custom experiments	100+ daily conversions	Not a fit
Google Conversion Lift	$5,000 minimum study	50 times
Meta Brand Lift	$15,000 default minimum	150 times

Want a comparison? Randomize after the click, on your own site.

It removes the audience problem,
not the noise. Still a hint at $100.

A test is a question with a bar

- One question. One page. One audience. One angle.
- Write your pass bar before you spend.
- Pass, fail, or unclear. There is no fourth answer.

The $100 threshold test

One of each

- **One** audience
- **One** landing page
- **One** conversion event: your signup
- **One** angle, one promise

Budget and timing

- $100 as a lifetime budget, with an end date
- 7 days
- Don't touch it unless it's broken
- Clean lowercase names and UTMs

The experiment card, 1 of 2

Fill it in and lock it before the first dollar.

01	**Question**	One sentence, tied to a decision you'll make.
02	**Hypothesis**	A claim with a number in it.
03	**One variable**	What you test, and everything you hold fixed.
04	**Audience and setup**	Platform, targeting, budget, dates, the signup event.

The experiment card, 2 of 2

01	**The bar**	PASS, FAIL, and UNCLEAR, written as signup counts.
02	**What you'll do**	Your next step for each of the three.
03	**Stop rules**	Broken tracking, a disapproved ad, overspend. Nothing else.
04	**Result**	Clicks, signups, rate, and the range. After the end date.

Commit it with a date. Don't edit it after launch.

From card to result

Lock
Fill in the card and commit it.

→

Prove
Sign up yourself. See it arrive.

→

Run
Submit early. 7 days, $100 lifetime.

→

Read
Once, after the end date.

Touch the ads only for a stop rule.

Then you do what the card said



PASS

Run a costlier second test: a pre-order, a price click, a booked call.



FAIL

Use the 60-second diagnosis. Change the promise or the page, not both.



UNCLEAR

Spend no more. Decide from interviews and other evidence.

A music ranking app on Meta

A hypothetical team, filled in from the research.

Question	Should we build signup-to-first-use for students who rank albums with friends?
Hypothesis	At least 5% of visitors (US students, 18 to 24, into music) join the waitlist.
One variable	The promise, Rank albums with friends. Page, audience, budget fixed.

Setup and the bar

Setup	Meta, Traffic objective, landing page views, one ad set, $100 lifetime, 7 days.
Tracking	Signups read from our own thank-you event. Tested with our own signup.
Expected	About 166 clicks, at about $0.60 a click.
The bar	PASS at 8 or more signups. FAIL at 7 or fewer.

PASS at 8 or more signups from 166 clicks.

A page truly at 2% passes 1.9% of the time.

A page truly at 8% passes 96% of the time.

Floor 2%, good 8%

Pick your bar

Clicks	PASS at	FAIL at	UNCLEAR
60	4+	1 or fewer	2 to 3
100	6+	3 or fewer	4 to 5
140	7+	5 or fewer	6
166	8+	7 or fewer	none

Each wrong call is held to about 5%.

Floor 3%, good 10%

Pick your bar

Clicks	PASS at	FAIL at	UNCLEAR
100	7+	4 or fewer	5 to 6
140	9+	7 or fewer	8
166	10+	9 or fewer	none

Floor 5%, good 10%

Clicks	PASS at	FAIL at	UNCLEAR
100	10+	4 or fewer	5 to 9
140	12+	7 or fewer	8 to 11
166	14+	9 or fewer	10 to 13

A wide gray band. $100 can't settle 5% against 10%.

Floor 5%, good 15%

Clicks	PASS at	FAIL at	UNCLEAR
100	10+	8 or fewer	9
140	12+	11 or fewer	none

Choose a bar two very different pages would land on opposite sides of.

If your real worry is 5% against 10%, $100 can't settle it.

8 signups from 166 clicks is 4.8%. The true rate: 2.5% to 9.2%.

Report a count and a range, not a rate alone.

166 clicks, as ranges

The range is where the true rate could sit.

Signups	Rate seen	Range for the true rate
0	0%	0% to 2.3%
3	1.8%	0.6% to 5.2%
8	4.8%	2.5% to 9.2%
17	10.2%	6.5% to 15.8%

Why you look only once

Two identical pages, 83 clicks each. How often does one falsely win?

- One look at the end: about 4 in 100
- Five looks, stop at the first win: about 10 in 100
- Ten looks: about 12 in 100

Check the dashboard only to see the ad runs.

After a PASS

- A PASS means worth a second, costlier test. Not "it works."
- Next, ask for more than an email: a pre-order, a price click, a booked call.
- Email or call the people who signed up. They tell you why.

Demand validation scorecard

Three numbers to read when the test ends.

CTR	Do they click? Measures attention.
CVR	Do they sign up? Measures promise.
Cost / signup	Is the signal affordable? Measures economics.

Clicks are not success. Signups are the only success.

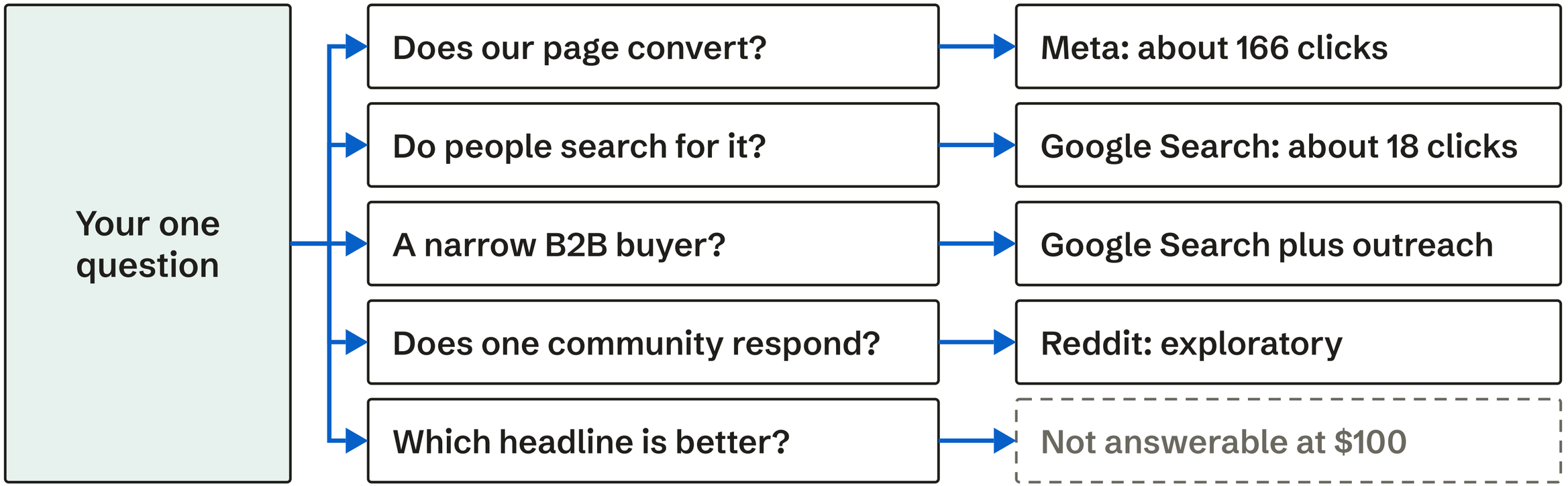
60-second diagnosis

Result	What it means	Fix
Low CTR	People aren't clicking	Hook, creative, or audience
Good CTR, low CVR	They click but don't sign up	Landing page clarity or friction
Good CVR, high cost	The signal is there	Creative efficiency or audience

Before touching ads, fix your hero and CTA first.

If the page doesn't convert, better ads just send more people to a broken page.

Pick the platform that answers your question



What each platform buys you

Which platform

	Meta	Google Search	Reddit
Clicks from $100	About 166	About 18	Unknown
Targets by	Interests, demographics	What people type	Communities, keywords
Ad review	Most within 24 hours	Most within 1 business day	Typically within a day
$100 answers	Does the page convert?	Is there search intent?	Does one community respond?

Your team's choice

If your team	Use	Why
Has a page and a signup event	Meta	Only one that reads a signup rate
Solves something people search for	Google Search	Tests intent and the words
Sells to a narrow B2B buyer	Google Search plus outreach	Replies matter more than clicks
Lives in a named community	Reddit, or Meta	Community targeting; costs unconfirmed

Pick one. Don't split $100 across platforms.

Each platform then gets too
few clicks to answer anything.

B2B teams: count people, not clicks

- $100 is about 17 Google clicks. You can't read a rate.
- Count booked calls and replies by hand.
- Message 20 to 30 target people yourself, next to the ads.
- Write the bar in people: 3 or more calls or real replies.

What fooled people in 21 Reddit threads

Too few clicks	Calling a result on about 60 clicks. Zero demos, nothing broken.
No tracking	Spending before anyone checked a signup actually registered.
The fix	Fill in your own form from the live ad.

fill in your own form from the live ad. if it registers, it's a tracking problem. if it doesn't, it's the page, and no bid tweak fixes that.

A commenter on r/PPC, July 2026

r/PPC: $626 spent on Google Search Ads, 61 clicks, zero leads

Two identical pages: one ends up 50% ahead 62 times in 100.

That's why you write the bar first and look once.

Studio: set up your ad and your card

Part 4

Your ad and your card

20:00

1. Pick the platform that answers your question.
2. Fill in your experiment card, then commit it.
3. Create the ad account: one owner, one card, two-factor on.
4. Check for verification or a spending limit.
5. **Hand in:** the committed card. Submit ads two business days early.

Before you launch, any platform

- One account owner, one card, two-factor on
- A thank-you page fires the signup event. Test it yourself.
- A lifetime or total budget with an end date, 7 days
- Submit ads 2 business days early

Landing page minimum

- A one-sentence promise that matches the ad
- 3 bullets: what you can do
- 1 mock or screenshot
- Email capture and a CTA

The ad promise and the page hero must match word for word.

Lowercase UTMs on every ad


```
utm_source=meta
&utm_medium=paid_social
&utm_campaign=teamname_product_nov2026
&utm_content=angle_a
```

Google Ads, 1 of 4: the account

- Create the account first, then build the campaign yourself
- Choose "Create a campaign without a goal's guidance"
- Check for a new-account daily spending limit
- Start advertiser verification early: up to 5 business days
- Don't count on a new-advertiser credit

Google Ads, 2 of 4: tracking

- A thank-you event that fires only after a real signup
- Mark the signup a key event in GA4
- Create the conversion under Goals, days ahead
- Test your own signup in GA4 Realtime

Google Ads, 3 of 4: the campaign

- Search, not Performance Max
- Maximize clicks, with a cost-per-click cap
- Campaign total budget: $100 over 7 days
- Turn AI Max off
- Untick Display; consider unticking search partners

Google Ads, 4 of 4: keywords and review

- 5 to 10 exact and phrase match phrases for the problem
- The ad repeats the page's promise word for word
- Set location and language
- Submit 2 business days early. Any edit restarts review.
- Don't use Google's Experiments feature

Meta, 1 of 2: account and tracking

- One owner of the ad account, the Page, and the card
- A business portfolio holding your Page and ad account
- Meta Pixel, with a Lead or CompleteRegistration event
- Read signups from your page or GA4, not Meta's Results

Meta, 2 of 2: the campaign

- Traffic objective, optimized for landing page views
- $100 lifetime budget with an end date, 7 days
- One ad set, one audience, one angle
- Don't edit mid-test. Submit 2 days early.
- Ignore the Learning label and the green trophy

Reddit Ads, exploratory

- Set a $100 campaign spend cap
- Skip Split Testing and Max campaigns
- Traffic objective, one targeting type
- Compare Reddit's clicks to your own visits
- Submit 48 hours early

Analytics: prove the tracking

Part 5

Two tools, two jobs

GA4: how they got to you

- Outside the product
- Traffic sources: Google, Reddit, direct
- Views, engagement time, engagement rate
- Free. Industry standard.

Amplitude: what they do

- Inside the product
- Custom events you instrument yourself
- Funnels, user paths, retention curves
- Free tier: 2M events a month

GA4 tells you how people arrived. Amplitude tells you what they did once they were there.

Building a mobile app?

Mobile

Amplitude has native iOS and Android SDKs: same events, same funnels. Firebase is Google's mobile SDK for GA4: apps report into the same Google Analytics.

GA4: the three reports that matter

01	**Acquisition**	Where are people coming from? This tells you which growth channels are working.
02	**Engagement**	How long are people staying? Are they bouncing immediately or exploring?
03	**Pages and screens**	Where do people land, and where do they leave? Your highest-exit page is your biggest leak.

Find these under Reports in GA4's left menu. Pages and screens is inside Engagement.

Amplitude: the three views that matter

01	**Event segmentation**	How many times is each event firing? This is your product heartbeat.
02	**Funnel analysis**	Define a multi-step flow. See where people drop off. This is your diagnostic tool.
03	**User paths**	What do users actually do after signup? Not what you designed. Often surprising.

Start from Create, then Chart: Event Segmentation, Funnel Analysis, or Journeys.

What to track

Defaults aren't enough

GA4 auto-tracks

- Page views, scroll depth, outbound clicks, site search, session start and end

Amplitude auto-tracks

- Session start and end, page views (if configured), app open and close (mobile), revenue events (if configured)

Defaults tell you someone visited, not what they did. You need custom events.

Custom events: map to your AARRR stages

Stage	What to track	Example events
Acquisition	Landing page visits, signup starts	page_view, signup_started
Activation	Signup completed, onboarding steps, first value action	signup_completed, onboarding_step_1, first_rating

Start with Acquisition and Activation events. Those are the stages you're in right now.

Custom events: the later stages

Stage	What to track	Example events
Retention	Return visits, repeat usage	session_start (day 2+), second_rating
Referral	Share actions, invite sends	invite_sent, profile_shared
Revenue	Purchase, subscription start	purchase_completed, trial_started

Start with 2 events, not 50

Amplitude's own documentation recommends starting with just two important events and expanding from there.

- Your signup event (Account Created)
- Your core value action (the thing that means "this user got it")

The signup event is what your ad test counts.

Name your events consistently

✓ Do this

- Object + Action, Title Case
- Account Created, Outfit Generated, Event Bookmarked
- Properties in snake_case (Segment's convention): referral_source, plan_type

✕ Not this

- click_button_3
- misc_event
- test123
- User Signed Up (2026-01-01)

Song Played and song played are two different events in Amplitude. Casing matters.

Your product has a critical path

Every product has a sequence of steps a user must complete to get value.



Step 1	Step 2	Step 3	Step 4	Step 5
Land on site	**Sign up**	**Onboarding**	**First value**	**Come back**

Each arrow is where people drop off. Find the biggest drop-off and fix it.

What to instrument

Track every step in each flow. Drop-off can happen anywhere.

Acquisition flow

Landing page viewed, CTA clicked, sign-up page viewed. Where do people bounce before they even try?

Sign-up flow

Email entered, password set, email verified, account created. "Sign-up completed" alone hides where they drop.

Activation flow

[First in-app action], [step 2 of your product], aha moment reached. Track every step of your aha journey.

Granularity is the point. Each step is a separate event.

If you only track "sign-up completed," you'll never know where in the flow people drop off.

Fire your first two events

Work time, 12 minutes

12:00

1. Name your signup event and your aha event, Object + Action.
2. Fire both from your product.
3. Sign up yourself, as a stranger would.
4. Watch the live event stream. See your signup arrive.
5. **Hand in:** a screenshot of your test signup arriving.

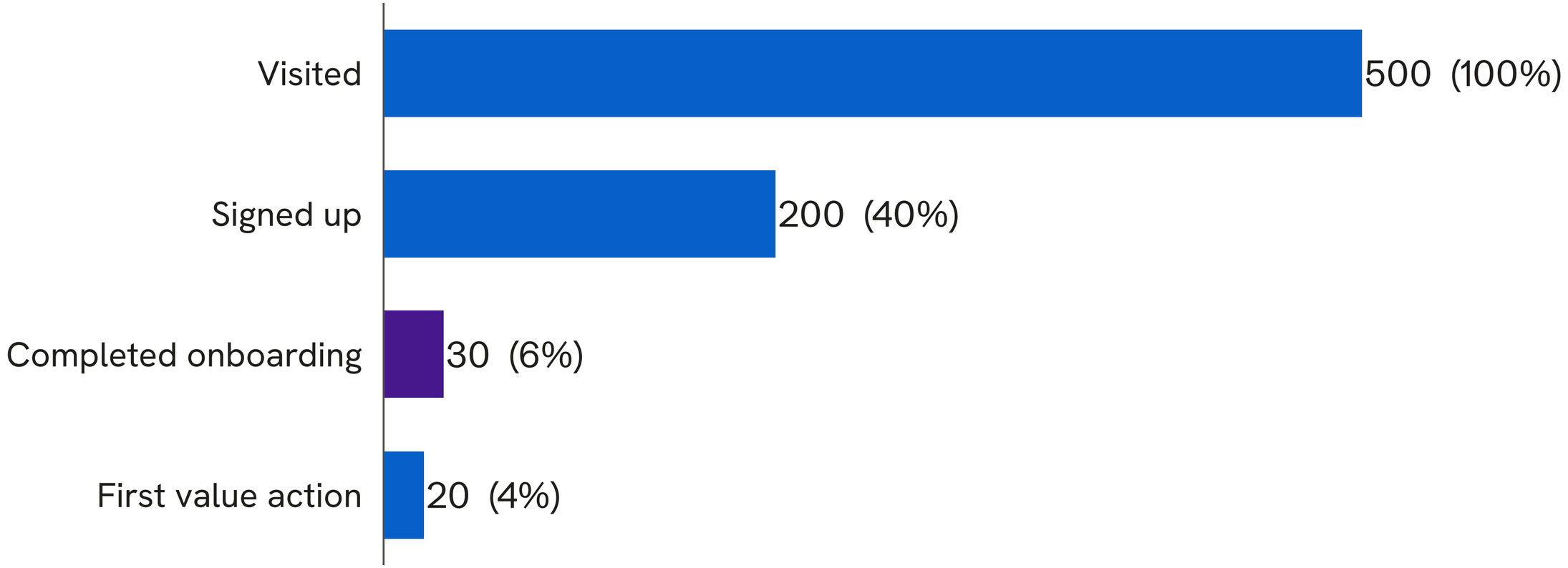
This week: get set up

01	**Install GA4**	Add the tracking snippet to your app. Set up conversion events for steps 1 to 3.
02	**Install Amplitude**	Add the SDK. Track steps 3 to 5 as Amplitude events. The free tier is enough.
03	**Get certified**	Complete Google Analytics for Beginners on Skillshop alongside your GA4 install.

You can't analyze data you haven't collected. Start this week.

How to read a funnel

Example: a funnel with a problem



The problem is obvious. 200 signed up but only 30 completed onboarding. 85% drop-off. Fix onboarding.

Example numbers

Each drop-off tells you something different

Different drop-offs, different fixes. The funnel tells you where to look.

01	**Visit to signup**	Landing page isn't compelling, or you're driving the wrong traffic.
02	**Signup to onboarding**	Onboarding is too confusing, too long, or asking too much too soon.
03	**Onboarding to value**	People finish onboarding but don't do the thing. The aha moment isn't clear.

Rough benchmarks (for orientation, not targets)

Funnel step	Typical range	Source
Landing page to signup	3-4% (SaaS median 3.8%)	Unbounce, 2024
Signup to activation	25-34%	Lenny's Newsletter, 2022
Day-7 retention, top 25%	7%	Amplitude, 2025
Free to paid, freemium	3-5% good, 8-12% great	Poyar and ChartMogul, 2026

With 30 users, benchmarks are context, not targets.

Your question is binary: are users reaching the aha moment, or aren't they?

If a number looks amazing, check the tracking before you celebrate.

Twyman's law, again. A double-firing signup event looks like a great week.

User testing and analytics

When you've watched 3 to 5 people use your product, you'll probably see them:

- Get confused at a specific step
- Click on things that don't work
- Miss the feature you thought was obvious
- Give up before reaching the aha moment

User testing and analytics

Both ways

These methods should reinforce each other. If you're seeing someone get confused, and you also see dropoff at that step of your funnel, then you know you've validated a problem in your product.

The clock is running

Until freeze

4

weeks of building time left before feature freeze, Friday, December 4

Demo Day

6

weeks to Demo Day, likely Friday, December 18

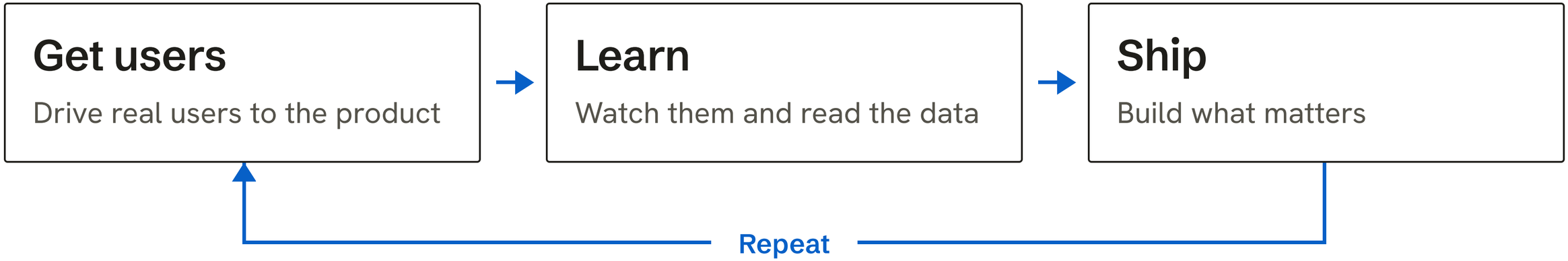
Starting

0

weeks to start getting users tonight

Everything from here is one loop

No more exploring. You have a product.



Find out if people can use it, then build the parts that make them stay.

Feature freeze: Friday, December 4.

Four weeks. After that: data collection and pitch prep only. The users you get NOW are the Demo Day data.

Homework

Due Friday, November 13, 1:10 PM.

This week: test, launch, and keep the plan current

Usability

3–5

sessions with real people, as class 8 assigned

Ad test

1

locked card, tracking proven, ad running

Growth plan

1

demand strategy, updated

Update

1

weekly update, before class

What to submit by Friday, November 13

- Usability tests: notes and a short write-up
- Tracking that works for ad traffic: Google Analytics tag, signup event, UTMs
- Your experiment card, locked, with the ad running
- Your demand strategy, updated
- Your weekly update

User testing: what to hand in

3 to 5 sessions on your onboarding flow. At least
1 stranger.

user_testing.md

What we observed: [pattern that appeared across sessions].
What we're changing: [specific product change]. Why: [what the sessions showed that led to this decision].

Tracking for your ad

1. The Google Analytics tag on the page your ad sends people to
2. Your signup event firing
3. UTMs on the ad's link
4. A screenshot of your own test signup arriving

Your ad experiment: what to hand in

- Your experiment card, sections 1 to 8, committed before your start date
- Proof of tracking: your test signup, arriving
- Your ads dashboard: campaign active, $100 lifetime, start and end dates
- The result goes in homework 9, due November 20

Your demand strategy, every Friday

- Homework 6's demand_strategy.md, moved to your team folder's top level
- Update it every Friday
- Your weekly update says what changed and why

Weekly update

Every team, before class every Friday.

- What we built last week and why
- What we're building this week and why
- Current user count
- Biggest blocker (if any)

Half a page. Not a novel.

Weekly update: the file


```
teams/your-team-name/weekly_update_YYYY-MM-DD.md
```

Homework

The and why is not optional.

Anyone can list tasks. I need to know whether you're deciding what to build and what to skip.

Next Friday: growth, cold start, and network effects

- Growth channels and outreach that gets read
- Why most network products die before they start
- Your cold start plan
- AstraPath and Settlewise present

Questions?

Write the bar before you spend

Next class · Friday, November 13, 2026 · 1:10 to 3:40 PM · Pupin 420